

Sikander Hayat Khan

Full Stack Developer (MongoDB · Express · React · Node.js · Next.js) · AI/ML Researcher · Agentic AI
National University of Sciences and Technology (NUST)
skhan.bese23seecs@seecs.edu.pk · +92 317 8598159 · Islamabad, Pakistan · [GITHUB](#) · [LINKEDIN](#) · [sikander-hayat-khan.vercel.app](#)

Professional Summary

Software Engineering student at NUST with hands-on experience across full-stack development, AI/ML research, and agentic AI systems. Proficient in the MERN stack and Next.js, with a track record of shipping production-quality applications. Co-authored a research paper on sentiment classification for a low-resource NLP task, and built a business research agentic AI system – demonstrating end-to-end ownership of complex technical problems.

Technical Skills

Frontend	React.js, Next.js, JavaScript (ES6+), HTML5, CSS3, Responsive Design, Figma
Backend	Node.js, Express.js, RESTful API Design, WebSocket (real-time), Flask (Python)
Database	MongoDB, PostgreSQL, Redis, MySQL, SQLite
AI/ML	PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, LoRA fine-tuning, Deep Neural Networks (DNN)
Agentic AI	LangGraph, LangChain, Groq (LLaMA 3), Tavily Search API, multi-agent pipelines, prompt engineering
Tools	Git, GitHub, Google Cloud (Storage, Compute), Kaggle (T4 GPU), Postman, CI/CD basics, Docker, Streamlit, Vercel, Render, Cloudinary

Research

Sentiment Analysis for Low-Resource NLP (Roman Urdu) | Co-Author · NUST SEecs (Islamabad · 2025)

- Co-authored a research paper tackling Roman Urdu sentiment classification – a 230M+ speaker language with no standardized orthography, annotated datasets, or task-specific NLP models.
- Designed and built a complete ML pipeline: Task-Adaptive Pre-Training (TAPT) on unlabeled Roman Urdu text, FGM adversarial training, Layer-wise Learning Rate Decay (LLRD), label smoothing, class-weighted loss, and a 4-seed soft-voting ensemble over XLM-RoBERTa.
- Achieved 80.53% accuracy and 80.62% weighted F1 on a 3-class sentiment task – a 42% relative improvement over the zero-shot baseline – using only free Kaggle T4 GPUs with no proprietary data.
- Also implemented LoRA ($r=8$, $\alpha=16$) fine-tuning adapting just ~0.3% of parameters, reaching 78.16% F1, validating the pipeline for resource-constrained environments.
- Python, PyTorch, HuggingFace Transformers, XLM-RoBERTa, LoRA, scikit-learn, Kaggle GPU**

Projects

Location-Based Journaling Platform | Web Engineering Project

(Islamabad · Nov 2025 – Jan 2026)

- Architected and built a full-stack web app with a React frontend and Node/Express backend, working in a team across design, backend, and QA – mirroring a real product workflow closely.
- Converted complex UX requirements into polished, responsive React components; implemented 8+ major product features including real-time chat, geotagged memory posts, map pins, calendar views, friend system, and client-side encrypted private memories.
- Designed RESTful APIs with CRUD, access-control middleware, WebSocket real-time messaging/live status, and a scalable cloud-integrated media upload pipeline.
- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, WebSockets, Cloud Storage, REST APIs

Search Engine for Medium Articles | DSA (Data Structures and Algorithms) Project

(Islamabad · Nov 2024 – Jan 2025)

- Built end-to-end over a 190k+ article dataset: scraping/cleaning, inverted index, BM25 multi-field ranking, a React frontend demo, and Flask API layer with Redis caching (median latency under 200ms).
- Integrated Google Gemini AI summarization, rich result cards, and smart filtering – iterated on relevance through A/B testing.
- **Tech Stack:** React (frontend), Python, Flask, BM25, Redis, Google Gemini API

Interactive Infectious Network Flow (IMF) App | DAA (Design and Analysis of Algorithms) Project

(Islamabad · Feb 2025 – May 2026)

- Built a real-time disease spread simulator over configurable community graphs with dynamic network evolution, D3.js-powered interactive visualization, and a multi-parameter disease model including vaccination and immunity mechanics.
- **Tech Stack:** React, TypeScript, D3.js, Tailwind CSS

Multi-Agent Business Research Assistant | AI/Agentic Systems · Personal Project

(Islamabad · May 2026)

- Built a 4-agent LangGraph pipeline (Clarity → Research → Validator → Synthesis) with conditional routing, Human-in-the-Loop interrupts, and MemorySaver checkpointing for multi-turn conversations.
- Integrated Tavily Search API for real-time web intelligence with self-scoring confidence evaluation, independent validation, and automatic retry logic to ensure production-grade research quality.
- **Tech Stack:** Python, LangGraph, LangChain, Groq (LLaMA 3.3-70b), Tavily Search API

Campus Pathfinder | Agentic AI/Students Support Triage System · Personal Project

(Islamabad · June 2026)

- Built an AI-powered student helpdesk triage agent (Groq/LLaMA 3) that automates intake-to-routing, reducing manual ticketing overhead by up to 80%.
 - Engineered a multi-stage LLM pipeline with automated issue classification, urgency scoring, smart follow-up prompting, and confidence-based department routing and escalation.
 - Delivered end-to-end ownership from AI layer to data layer: auto-generated ticket summaries, staff response email drafting, and full SQLite persistence with CRUD support.
 - **Tech Stack:** Python, Streamlit, Groq (LLaMA 3), SQLite, LLM Prompt Engineering
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Education

Bachelor of Engineering (B.E.) in Software Engineering

National University of Sciences and Technology (NUST), Islamabad

Sep 2023 – Jun 2027 (Expected)

Relevant coursework: Data Structures & Algorithms, Database Systems, Web Engineering, Computer Networks, Software Development & Architecture, Machine Learning, Software Quality Engineering.

Developed academic projects using React, Node.js, Express.js, PostgreSQL, REST APIs, and modern software development practices.

Higher Secondary School Certificate (HSSC)

Abbottabad Public School (APS), Abbottabad

May 2015 – May 2021

Pre-Engineering

Certifications

Supervised Machine Learning: Regression & Classification – DeepLearning.AI & Stanford (Coursera) · Jul 2025

Advanced Learning Algorithms – DeepLearning.AI & Stanford (Coursera) · Aug 2025

Google Cloud: Cloud Storage & Compute Skill Badges – Google Cloud · Oct 2025

Leadership & Activities

- ▶ Led a team of 8 in NUST stage drama – negotiated vendor contracts, managed set design, lighting, and scheduling for two 30-min performances with zero cancellations.
- ▶ Organized campus-wide Ramadan Ration Drive (Feb–Mar 2026): fundraising, volunteer management, and distribution to underprivileged families.
- ▶ Futsal Champions – SEECS Sports Gala, NUST (Oct–Nov 2023).
- ▶ Led the Finance Team for **Khaapa** – an entrepreneurship venture at SEECS, NUST where we set up a student-run campus shop selling snacks, meals, and beverages; managed revenue tracking, cost control, and budgeting across the full sales cycle.